Catherine **Cerny**

University of Michigan, Department of Astronomy. 1085 S. University Ann Arbor, MI 48109

(517) 862-2977 | cecerny@umich.edu | cecerny.github.io

Education

PhD - Physics — *Durham, UK*
Durham University — *Oct. 2020 - July 2024*

Master of Arts - Astronomy — *Boston, MA*
Boston University — *Sept. 2017 - Aug. 2019*

Bachelor of Science - Highest Honors in Astronomy and Astrophysics — *Ann Arbor, MI*
University of Michigan — *Sept. 2012 - Dec. 2016*

Bachelor of Science - Interdisciplinary Physics — *Ann Arbor, MI*
University of Michigan — *Sept. 2012 - Dec. 2016*

Bachelor of Arts - Asian Studies — *Ann Arbor, MI*
University of Michigan — *Sept. 2012 - Dec. 2016*

Research Experience

Postdoctoral Work

"Dark Matter and Strong Gravitational Lensing" — *University of Michigan*
University of Michigan; Supervisor: Professor Keren Sharon — *Sept. 2024 - present*

- Led development of a strong gravitational lens model for a complex exotic configuration of lensed galaxies in the galaxy cluster RXJ0437, with a particular focus on accounting for statistical biases within the modeling process that influence the detection of small-scale dark matter substructures (**Cerny et al. 2026b**, submitted to ApJ)
- Supervised and mentored an undergraduate student, which led to the student's first publication (**Smith et al. 2025**)
- Serve as a lens modeling expert in the LEnsing and Galaxy Growth: Observing Substructures (LEGGOS) collaboration; (**Abedi et al. 2026**, **Khullar et al. 2026**)

"Strong LensIng and Cluster Evolution (SLICE) Across Cosmic Time" — *University of Michigan*
University of Michigan; Supervisor: Professor Keren Sharon — *Sept. 2024 - present*

- Serve as the co-lead of the strong lensing working group for the Strong LensIng and Cluster Evolution (SLICE) JWST program
- Led an international team of collaboration members to create the first publication for SLICE (**Cerny et al. 2026a**)
- Oversee lens modeling efforts for 124 galaxy clusters through the analysis of JWST/HST imaging, VLT/MUSE spectroscopy, Chandra X-ray observations, and other archival data
- Co-leading a data release effort that identifies all lensed sources, measures spectroscopic redshifts, and presents first-order lens models for nearly one hundred galaxy clusters in SLICE (**Cerny, Beauchesne et al., in prep.**)
- Manage JWST, HST, VLT/MUSE, ancillary archival data for 124 clusters in SLICE
  - Perform data reduction and processing of archival HST imaging and VLT/MUSE datacubes
  - Developed and deployed organizational spreadsheets and automated scripts to facilitate access to data products
- Direct the observation of ground-based targeting for lensed objects in SLICE with Magellan/FIRE and Magellan/LDSS3, including on-site observing experience at the Las Campanas Observatory in Chile

PhD Work

"Examining the Core-Cusp Problem with Strong Gravitational Lensing " *Durham, UK*

Durham University; Advisor: Professor Mathilde Jauzac *Oct. 2020 - July 2024*

- Placed constraints on the shape of the dark matter density profile in clusters by using multi-probe analysis, developing new techniques for using VLT/MUSE spectroscopy in lens modeling to constrain BCG masses (**Cerny et al. 2025**)
- Analyzed VLT/MUSE datacubes to extract spectroscopic redshift measurements of galaxy cluster members and lensed objects for the Kaleidoscope survey (**Edge et al.**, in prep.)

Masters Work

"Strong Lensing Analysis of the Merging Galaxy Cluster SGAS J122651.3+215 for JWST" *Ann Arbor, MA*

University of Michigan; Advisor: Professor Keren Sharon *Sept. 2019 - Oct. 2020*

- Developed lens model for the galaxy cluster SGAS J1226 in preparation for TEMPLATES observations of its bright, highly magnified arc with JWST (**Sharon, Cerny et al. 2023**)

"B-Field Tomography in the Milky Way Using Stellar Distances from *Gaia*" *Boston, MA*

Boston University; Advisor: Dr. Dan Clemens *May 2017 - May 2019*

- Analyzed *Gaia* data to characterize the 3D magnetic field structure of the Milky Way in the Galactic Plane Infrared Polarization Survey (**(Clemens, El-Batal, Cerny et al 2018**; **Clemens, Cashman, Cerny et al. 2020**)

Undergraduate Work

"Strong Lens Models for Massive Galaxy Clusters in the Reionization Lensing Cluster Survey" *Ann Arbor, MI*

University of Michigan; Advisor: Dr. Keren Sharon *Sept. 2015 – August 2017*

- Developed lens models for galaxy clusters based on strong gravitational lensing in order to search for galaxies at z 9-12 as a part of the RELICS collaboration
- Led the publication of the first paper from the RELICS program in ApJ (**Cerny et al. 2018**)

Presentations

Conferences **2 Invited Talks - 21 Talks**

Sept. 25, 2026	**Talk**, Lenstool Days	*Grenoble, France*
Aug. 11, 2026	**Talk**, JWST Through Cluster Lenses	*Santander, Spain*
Oct. 20, 2025	**Talk**, Lenstool Days	*Liege, Belgium*
June 27, 2025	**Invited Talk**, EAS Annual Meeting 2025	*Cork, Ireland*
Oct. 1, 2024	**Talk**, SPT Collaboration Meeting	*Chicago, Illinois*
July 13, 2023	**Invited Talk**, EAS Annual Meeting 2023	*Krakow, Poland*
Oct. 5, 2023	**Talk**, Cluster Physics From Space To Reveal Dark Matter	*Bern, Switzerland*
Mar. 6, 2023	**Talk**, Cosmology on Safari 2023	*Hluhluwe, KwaZulu-Natal, South Africa*
Jan. 10, 2023	**Talk**, DEX-XIX 2023 eXtragalactic Workshop	*Edinburgh, UK*
Nov. 26, 2022	**Talk**, Conference for Women in Physics	*Karlsruhe, Germany*
Jul. 21, 2022	**Talk**, Cluster Physics From Space To Reveal Dark Matter	*Bern, Switzerland*
Jun. 28, 2022	**Talk**, EAS Annual Meeting 2022	*Valencia, Spain*

Apr. 27, 2022	**Talk**, Galaxy Clusters 2022	*Space Telescope Science Institute, US*
Apr. 5, 2022	**Talk**, BUFFALO Collaboration Meeting	*Marseille, France*
Jan. 6, 2022	**Talk**, DEX-XVIII 2022 Virtual Conference	*Durham, UK*
Jul. 22, 2021	**Talk**, National Astronomy Meeting NAM2021	*University of Bath, UK*
Jul. 12, 2021	**Talk**, BUFFALO Virtual Collaboration Meeting	*Durham University, UK*
Jul. 6, 2021	**Talk**, Lenstool Days	*CRAL (Lyon), France*
Jul. 2, 2021	**Talk**, EAS Annual Meeting 2021	*Leiden, The Netherlands*
Jan. 6, 2021	**Talk**, DEX-XVII 2021 Virtual Conference	*Edinburgh, UK*
Jan. 2017	**Talk**, 229th AAS Meeting	*Grapevine, TX*

Local Presentations

Jun. 3, 2025	**Talk**, Laboratoire d'Astrophysique de Marseille café-club	*Marseille, France*
Jan. 16, 2025	**Guest Lecture**, University of Michigan Astronomy 220	*Ann Arbor, Michigan*
Sept. 19, 2024	**Talk**, University of Michigan Department of Astronomy Colloquium	*Ann Arbor, Michigan*
Nov. 4, 2022	**Talk**, Durham University Friday Afternoon Lunch Talk	*Durham, UK*
May 21, 2021	**Talk**, Durham University Friday Afternoon Lunch Talk	*Durham, UK*
Jan. 2019	**Talk**, Boston University Astronomy Department Student Seminar	*Boston, MA*
Aug. 2016	**Talk**, Honors Summer Fellowship Presentation Session	*Ann Arbor, MI*

Committees

2024 - present	**Co-Lead of Strong Lensing Working Group**, SLICE Collaboration	*Ann Arbor, MI*
2023	**Organizer & Session Chair**, NAM 2023, "Probing dark matter and baryons with galaxy clusters"	*Cardiff, UK*

Honors and Awards

2023 Scientific High Level Visiting Fellowship

Marseille, France

Ambassade de France au Royaume-Uni

September 2023

- Awarded grant to support one month in-person collaboration at the Laboratoire d'Astrophysique de Marseille with Dr. Eric Jullo and Dr. Marceau Limousin

Durham University Studentship (Faculty of Science)

Durham, UK

Durham University

Oct. 2020 - March 2024

- Full scholarship to support postgraduate studies at Durham University

Visiting Intern Fellowship

Boston, MA

Harvard University

May 2019 - Aug. 2019

- Awarded fellowship to support summer research at the Center for Astrophysics

Dean's Fellowship

Boston, MA

Boston University

Sept. 2017 - May 2018

- Awarded full fellowship to support first year of study at Boston University

Outreach and Service

Postdoctoral Representative
Ann Arbor, MI

VOLUNTEER
Sept. 2024 - present

- Serve as one of two representatives for postdocs in the University of Michigan's Department of Astronomy faculty meetings

University of Michigan Telescope Allocation Commiteee
Ann Arbor, MI

COMMITTEE MEMBER
November 2025

- Served as a member of the internal University of Michigan Astronomy department Magellan/MDM TAC for the 2026A semester

Department of Astronomy Social Committee
Durham, UK

COMMITTEE MEMBER
Oct. 2020 - Oct. 2023

- Planned and organized social events for astronomy department

Astronomy Open Night
Boston, MA

VOLUNTEER
Sep. 2017 - May 2019

- Participated in Department of Astronomy outreach program designed to introduce the general public to observational astronomy

Girls Science Club
Boston, MA

VOLUNTEER
Sep. 2018 - May 2019

- Introduced elementary school girls to basic science concepts by creating and presenting interactive projects to encourage participation and curiosity

Physics Tutoring
Boston, MA

TUTOR
Sep. 2017 - May 2018

- Mentored introductory physics students to help them become comfortable with using basic physics concepts in class

Student Astronomical Society
University of Michigan, Ann Arbor, MI

MEMBER
Dec. 2015 - Dec. 2016

- Hosted bi-weekly open houses during the semester, which included planetarium shows, computer-based activities, and outdoor observing geared toward public outreach

Skills

Software Skills	Reduce and analyze JWST, HST, VLT/MUSE data
	Photometric analysis of JWST and HST lensed sources using Photutils
	Write scripts for and Python implementations of processes executed with MCMC algorithm `Lenstool`
	Data reduction of *Chandra* data with CIAO
	SQL database management of large amounts of *Gaia* data
Observing Experience	Remotely observed with Magellan/FIRE and Magellan/LDSS3 for a total of 18 nights
	Operated the Magellan telescopes at Las Campanas Observatory for two nights
	Operated the MDM 1.3m McGraw-Hill telescope for two nights
	Operated the Perkins 1.8m telescope for one night
Programming	Python (matplotlib, pandas, astropy, arviz), SQL, Google Apps Script
System Management	Compiling and installing software packages for both off the shelf and custom distributions
	Developing and deploying Google spreadsheets and information databases for SLICE
Languages	English (native speaker)
	Japanese (read, spoken, written)